

Second Order System

Quadratic lag $\rightarrow$ Second order differential eqn

Can be series of first order systems / Addition of controllers to first order system

OR

OK
Inherently second order systems

$$A \frac{d^2 y}{dt^2} + B \frac{dy}{dt} + C y = x(t)$$

Manometer

Initially Pressure same on both legs
Length of fluid = L

@ $t=0$; Pressure differential applied

fluid moves in Laminar flow

Transfer function between applied pressure ΔP
and reading on manometer

Momentum Balance:-

Sum of forces
causing fluid
motion = Rate of change
of fluid
momentum

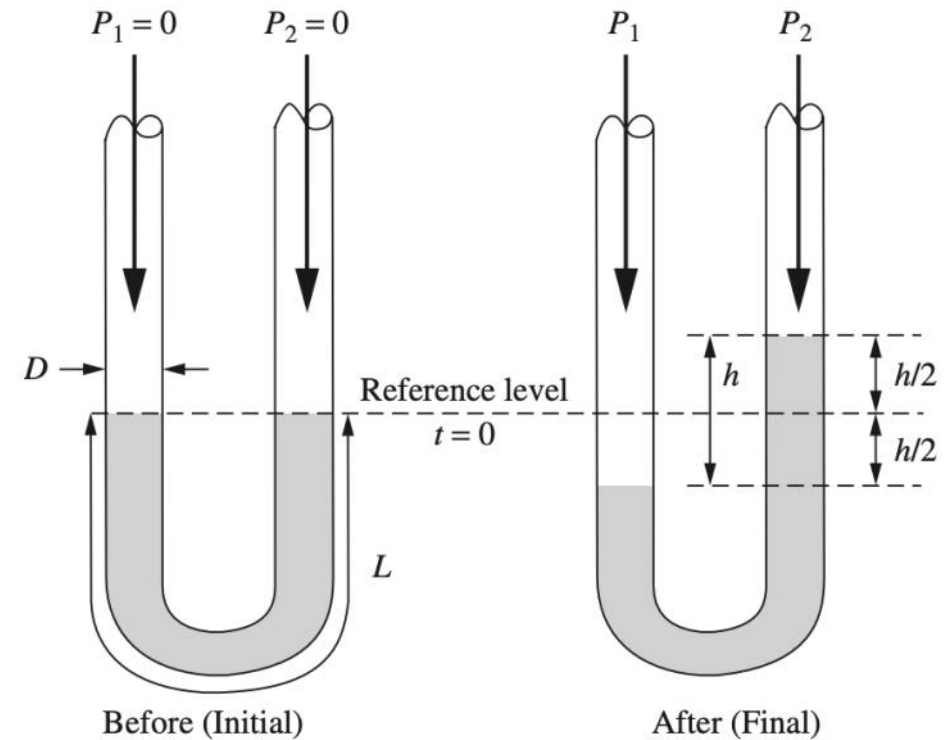


FIGURE 7-1
Manometer.

Momentum Balance-

Sum of forces causing fluid to move = Rate of change of momentum of fluid

Unbalanced pressure forces causing motion

$$(P_1 - P_2) \frac{\pi D^2}{4} - \rho g h \frac{\pi D^2}{4}$$

- frictional forces resisting motion

Shear stress at wall τ_{wall} $\left(\begin{array}{l} \text{Skin friction} \\ \text{Area in contact with wall} \end{array} \right)$

$$\tau_{wall} (\pi D L)$$

$$= \frac{8 \mu \bar{V}}{D} (\pi D L)$$

$$= \frac{8 \mu}{D} \left(\frac{1}{2} \frac{dh}{dt} \right) (\pi D L)$$

$$\bar{V} \rightarrow \text{avg velocity of fluid}$$

$$= \text{velocity of interface}$$

$$= \frac{1}{2} \frac{dh}{dt}$$

$$\frac{d}{dt} (\text{mass} \times \text{velocity} \times \text{correction})$$

$$= \left(\rho \frac{\pi D^2}{4} L \right) \frac{d\bar{V}}{dt} \beta \rightarrow \text{Parabolic Path correction}$$

$$\hookrightarrow \text{Rate of change of velocity}$$

$$= \left(\rho \frac{\pi D^2}{4} L \right) \left(\frac{1}{2} \frac{d^2 h}{dt^2} \right) \beta \left| \frac{4}{3} \text{ @ Laminar flow} \right.$$

$$\left\{ (P_1 - P_2) \frac{\pi D^2}{4} - \rho g h \frac{\pi D^2}{4} \right\} - \left\{ \frac{8\mu}{D} \frac{1}{2} \frac{dh}{dt} \pi D L \right\} = \left(\rho \frac{\pi D^2}{4} L \right) \left(\frac{1}{2} \frac{d^2 h}{dt^2} \right) \left(\frac{4}{3} \right)$$

Rearranging as second order eqn

$$\frac{4}{\rho g \pi D^2} \left\{ \left(\rho \frac{\pi D^2}{4} L \right) \left(\frac{4}{3} \right) \left(\frac{1}{2} \frac{d^2 h}{dt^2} \right) + \left(\frac{8\mu}{D} \right) \left(\frac{1}{2} \frac{dh}{dt} \right) (\pi D L) + \rho g h \frac{\pi D^2}{4} \right\} = (P_1 - P_2) \frac{\pi D^2}{4}$$

$$\underbrace{\frac{2L}{3g}}_{\tau^2} \frac{d^2 h}{dt^2} + \underbrace{\frac{16\mu L}{\rho g D^2}}_{2\zeta\tau} \frac{dh}{dt} + \underbrace{h}_{Y} = \underbrace{\frac{\Delta P}{\rho g}}_{X(t)}$$

$$\tau = \sqrt{\frac{2L}{3g}}$$

$$\zeta = \frac{8\mu}{\rho D^2} \sqrt{\frac{2L}{3g}}$$

$$\tau^2 \frac{d^2 h}{dt^2} + 2\zeta\tau \frac{dh}{dt} + Y = X(t)$$

$$\Rightarrow 2\zeta\tau = 2 \frac{8\mu}{\rho D^2} \sqrt{\frac{2L}{3g}} \sqrt{\frac{32}{2g}} = \frac{16\mu L}{\rho D^2 g}$$

Laplace Transform

$$\tau^2 s^2 Y(s) + 2\zeta\tau s Y(s) + Y(s) = X(s) \Rightarrow$$

$$\boxed{\frac{Y(s)}{X(s)} = \frac{1}{\tau^2 s^2 + 2\zeta\tau s + 1}}$$

Step Response

$$X(s) = \frac{1}{s}$$

$$Y(s) = \frac{1}{s} \frac{1}{\tau^2 s^2 + 2\zeta\tau s + 1} \implies$$

Finding roots

$$= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-2\zeta\tau \pm \sqrt{(2\zeta\tau)^2 - 4(\tau^2)(1)}}{2\tau^2}$$

$$= \frac{-\cancel{\tau}\cancel{\tau} \pm \frac{2\tau^2}{\sqrt{4\zeta^2\tau^2 - 4\tau^2}}}{\cancel{\tau}\cancel{\tau^2}}$$

$$= -\frac{\zeta}{\tau} \pm \frac{\sqrt{4\tau^2(\zeta^2 - 1)}}{2\tau^2}$$

$$= -\frac{\zeta}{\tau} \pm \frac{\sqrt{\zeta^2 - 1}}{\tau}$$

Factors are of the form
 $a(s - s_a)(s - s_b)$

$$s_a = -\frac{\zeta}{\tau} - \frac{\sqrt{\zeta^2 - 1}}{\tau}, \quad s_b = -\frac{\zeta}{\tau} + \frac{\sqrt{\zeta^2 - 1}}{\tau}$$

$$Y(s) = \frac{1}{a s(s - s_a)(s - s_b)}$$

$$Y(s) = \frac{1/\tau^2}{s(s - s_a)(s - s_b)}$$

s_a and s_b are real or complex
 based on value of ζ
 $\hookrightarrow$ The nature of root will
 effect $y(t)$

Three Possible Cases

$\zeta < 1 \rightarrow$ Complex roots $\rightarrow$ Underdamped / Oscillatory system

$\zeta = 1 \rightarrow$ Real and equal roots $\rightarrow$ Critically damped

$\zeta > 1 \rightarrow$ Real roots $\rightarrow$ Over damped / Non Oscillatory system

Case 1
for $\zeta < 1$ under damping
and a Step change

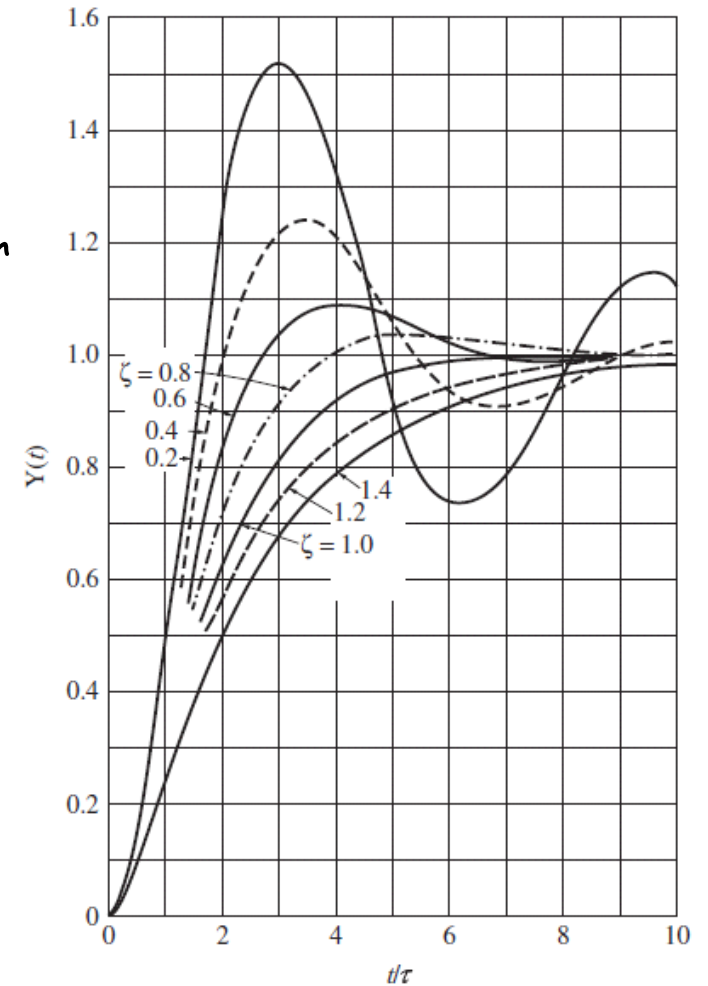


FIGURE 7-3

Response of a second-order system to a unit-step forcing function.

$$Y(s) = \frac{1}{s(\tau^2 s^2 + 2\zeta\tau s + 1)} = \frac{A}{s} + \frac{Bs + C}{\tau^2 s^2 + 2\zeta\tau s + 1}$$

$$1 = A(\tau^2 s^2 + 2\zeta\tau s + 1) + (Bs + C)s$$

$$@ s=0 ; A=1$$

$$\Rightarrow \tau^2 s^2 + 2\zeta\tau s + 1 + Bs^2 + Cs = 1$$

$$(\tau^2 + B)s^2 + (2\zeta\tau + C)s + 1 = 1$$

$$\hookrightarrow \tau^2 + B = 0 ; 2\zeta\tau + C = 0$$

$$B = -\tau^2 ; C = -2\zeta\tau$$

$$\Rightarrow Y(s) = \frac{1}{s} - \frac{\tau^2 s + 2\zeta\tau}{\tau^2 s^2 + 2\zeta\tau s + 1}$$

for perfect square; quadratic form
 $s^2 + as + (\frac{a}{2})^2 = (s + \frac{a}{2})^2$

$\Rightarrow$ For

$$\frac{\tau^2 s + 2\zeta\tau}{\tau^2 s^2 + 2\zeta\tau s + 1}$$

$$\hookrightarrow \tau^2 \left(s^2 + \frac{2\zeta\tau s}{\tau^2} + \frac{1}{\tau^2} \right)$$

$$= \tau^2 \left(\left(s + \frac{\zeta\tau}{\tau^2} \right)^2 + \frac{1}{\tau^2} - \frac{\zeta^2}{\tau^2} \right)$$

$$= \tau^2 \left(\left(s + \frac{\zeta}{\tau} \right)^2 + \frac{1}{\tau^2} - \frac{\zeta^2}{\tau^2} \right)$$

$$= \tau^2 \left(s + \frac{\zeta}{\tau} \right)^2 + (1 - \zeta^2)$$

$$= \frac{\tau^2 s + 2\zeta\tau}{\tau^2 \left(s + \frac{\zeta}{\tau} \right)^2 + (1 - \zeta^2)}$$

$$Y(s) = \frac{1}{s} - \frac{s^2 + 2\zeta s}{s^2 \left(s + \frac{\zeta}{2}\right)^2 + \left(1 - \frac{\zeta^2}{4}\right)}$$

$$= \frac{1}{s} - \frac{s + \frac{2\zeta}{2}}{\left(s + \frac{\zeta}{2}\right)^2 + \left(\frac{1}{2^2} - \frac{\zeta^2}{2^2}\right)}$$

$$= \frac{1}{s} - \frac{\left(s + \frac{\zeta}{2}\right) + \frac{\zeta}{2}}{\left(s + \frac{\zeta}{2}\right)^2 + \left(\frac{1}{2^2} - \frac{\zeta^2}{2^2}\right)}$$

$$= \frac{1}{s} - \frac{\left(s + \frac{\zeta}{2}\right)}{\left(s + \frac{\zeta}{2}\right)^2 + \left(\frac{1 - \zeta^2}{2^2}\right)} - \frac{\frac{\zeta}{2}}{\left(s + \frac{\zeta}{2}\right)^2 + \left(\frac{1 - \zeta^2}{2^2}\right)}$$

$$\mathcal{L}\{e^{-at} \sin(kt)\} = \frac{k}{(s+a)^2 + k^2}$$

$$\mathcal{L}\{e^{-at} \cos(kt)\} = \frac{(s+a)}{(s+a)^2 + k^2}$$

$$Y(s) = \frac{1}{s} - \frac{\left(s + \frac{\zeta}{2}\right)}{\left(s + \frac{\zeta}{2}\right)^2 + \left(\frac{\sqrt{1-\zeta^2}}{2}\right)^2} - \frac{\frac{\zeta}{2}}{\left(s + \frac{\zeta}{2}\right)^2 + \left(\frac{\sqrt{1-\zeta^2}}{2}\right)^2} \rightarrow \frac{\zeta}{2} = \frac{\zeta}{2} \frac{\sqrt{1-\zeta^2}}{\sqrt{1-\zeta^2}} = \left(\frac{\sqrt{1-\zeta^2}}{2}\right) \left(\frac{\zeta}{\sqrt{1-\zeta^2}}\right)$$

$$\mathcal{L}^{-1} \left[\frac{s+a}{(s+a)^2 + b^2} \right] = e^{-at} \cos(bt)$$

$$\mathcal{L}^{-1} \left[\frac{b}{(s+a)^2 + b^2} \right] = e^{-at} \sin(bt)$$

$$\begin{aligned} Y(t) &= 1 - e^{-\frac{\zeta}{2}t} \cos\left(\frac{\sqrt{1-\zeta^2}}{2}t\right) - \frac{\zeta}{\sqrt{1-\zeta^2}} e^{-\frac{\zeta}{2}t} \sin\left(\frac{\sqrt{1-\zeta^2}}{2}t\right) \\ &= C_1 + e^{-\frac{\zeta}{2}t} \left(C_2 \cos \frac{\sqrt{1-\zeta^2}}{2}t + C_3 \sin \frac{\sqrt{1-\zeta^2}}{2}t \right) \Rightarrow \end{aligned}$$

Using identity

$$A \cos \theta + B \sin \theta = R \sin(\theta + \phi)$$

$$\text{with } R = \sqrt{A^2 + B^2} \quad \text{and } \tan \phi = \frac{A}{B}$$

$$A = -1, \quad B = -\frac{\zeta}{\sqrt{1-\zeta^2}}$$

$$R = \sqrt{1 + \frac{\zeta^2}{1-\zeta^2}} = \sqrt{\frac{1-\zeta^2 + \zeta^2}{1-\zeta^2}} = \frac{1}{\sqrt{1-\zeta^2}}$$

$$\text{and } \tan \phi = \frac{A}{B} = \frac{-1}{\frac{-\zeta}{\sqrt{1-\zeta^2}}} = \frac{\sqrt{1-\zeta^2}}{\zeta} \Rightarrow \phi = \tan^{-1} \frac{\sqrt{1-\zeta^2}}{\zeta}$$

$$\text{Finally } y(t) = 1 - R e^{-\zeta t / \tau} \sin\left(\frac{\sqrt{1-\zeta^2}}{\tau} t + \phi\right) = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta t / \tau} \sin\left(\frac{\sqrt{1-\zeta^2}}{\tau} t + \tan^{-1} \frac{\sqrt{1-\zeta^2}}{\zeta}\right)$$

Case II

$\zeta = 1$ Critical damping

as roots s_a and s_b are

$$s_a = -\frac{\zeta}{2} - \frac{\sqrt{\zeta^2 - 1}}{2}, \quad s_b = -\frac{\zeta}{2} + \frac{\sqrt{\zeta^2 - 1}}{2}$$

$$s_a = s_b = -\frac{1}{2}$$

$$Y(s) = \frac{\frac{1}{\tau^2}}{s(s + \frac{1}{\tau})(s + \frac{1}{\tau})} = \frac{\frac{1}{\tau^2}}{s(s + \frac{1}{\tau})^2} = \frac{A}{s} + \frac{B}{(s + \frac{1}{\tau})} + \frac{C}{(s + \frac{1}{\tau})^2}$$

$$\frac{1}{\tau^2} = A(s + \frac{1}{\tau})^2 + Bs(s + \frac{1}{\tau}) + Cs$$

$$\frac{1}{\tau^2} = A\left(s^2 + \frac{1}{\tau^2} + \frac{2s}{\tau}\right) + Bs^2 + B\frac{s}{\tau} + Cs$$

Comparing terms

$$s^2 \Rightarrow A + B = 0$$

$$1 + B = 0 \\ B = -1$$

$$; s \Rightarrow A \frac{2}{s} + \frac{B}{s} + C = 0 \Rightarrow \frac{1}{s^2} = \frac{A}{s^2} \Rightarrow A = 1$$

$$\frac{2}{s} - \frac{1}{s} + C = 0$$

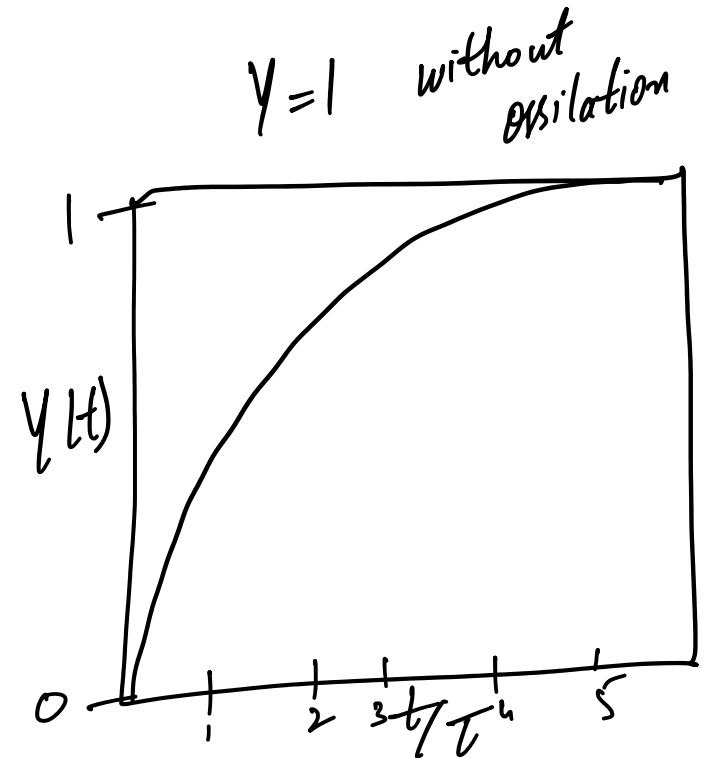
$$C = -\frac{2-1}{s} = -\frac{1}{s}$$

Hence

$$\frac{\frac{1}{s^2}}{s(s + \frac{1}{2})^2} = \frac{1}{s} - \frac{1}{(s + \frac{1}{2})} - \frac{\frac{1}{2}}{(s + \frac{1}{2})^2}$$

$\downarrow$ $\downarrow$ $\downarrow$
 1 $e^{-t/2}$ $\frac{t}{2} e^{-t/2}$
Inverse

$$y(t) = 1 - e^{-t/2} - \frac{t}{2} e^{-t/2} = 1 - \left(1 + \frac{t}{2}\right) e^{-t/2}$$



Case III

$$\zeta > 1$$

Over damping

$$Y(s) = \frac{1}{s} \frac{1}{(s-s_a)(s-s_b)} = \frac{A}{s} + \frac{B}{(s-s_a)} + \frac{C}{(s-s_b)}$$

$$1 = A(s-s_a)(s-s_b) + Bs(s-s_b) + Cs(s-s_a)$$

$$@ s=0 \Rightarrow A = \frac{1}{s_a s_b}$$

$$@ s=s_a \Rightarrow 1 = B s_a (s_a - s_b) \Rightarrow B = \frac{1}{s_a (s_a - s_b)}$$

$$@ s=s_b \Rightarrow C = \frac{1}{s_b (s_b - s_a)}$$

$$Y(s) = \frac{1}{s_a s_b} \frac{1}{s} + \frac{1}{s_a (s_a - s_b)} \frac{1}{(s-s_a)} + \frac{1}{s_b (s_b - s_a)} \frac{1}{(s-s_b)}$$

$$Y(t) = \frac{1}{s_a s_b} + \frac{e^{s_a t}}{s_a (s_a - s_b)} + \frac{e^{s_b t}}{s_b (s_b - s_a)} = 1 + \frac{s_b e^{s_a t}}{(s_a - s_b)} - \frac{s_a e^{s_b t}}{(s_a - s_b)}$$

$$\text{Solving } s_a - s_b = -\frac{\zeta}{\cancel{2}} + \frac{\sqrt{\zeta^2 - 1}}{2} - \left(-\frac{\zeta}{\cancel{2}} - \frac{\sqrt{\zeta^2 - 1}}{2} \right) = 2 \frac{\sqrt{\zeta^2 - 1}}{2}$$

$$\text{Solving } s_a + s_b = -\frac{\zeta}{2} + \frac{\sqrt{\zeta^2 - 1}}{2} + \left(-\frac{\zeta}{2} - \frac{\sqrt{\zeta^2 - 1}}{2} \right) = -2 \frac{\zeta}{2}$$

$$\begin{aligned} \text{and } s_b e^{s_a t} - s_a e^{s_b t} \\ = \frac{1}{2} \left[(-\zeta - \sqrt{\zeta^2 - 1}) e^{\frac{(-\zeta + \sqrt{\zeta^2 - 1})}{2} t} - (-\zeta + \sqrt{\zeta^2 - 1}) e^{\frac{(-\zeta - \sqrt{\zeta^2 - 1})}{2} t} \right] \\ = e^{-\frac{\zeta}{2} t} \left[s_b e^{\frac{\sqrt{\zeta^2 - 1}}{2} t} - s_a e^{-\frac{\sqrt{\zeta^2 - 1}}{2} t} \right] \end{aligned}$$

$$\text{Using } \sinh a = \frac{e^a - e^{-a}}{2} \quad ; \quad \cosh a = \frac{e^a + e^{-a}}{2}$$

$$e^{\frac{\sqrt{\zeta^2 - 1}}{2} t} - e^{-\frac{\sqrt{\zeta^2 - 1}}{2} t} = 2 \sinh \left(\frac{\sqrt{\zeta^2 - 1}}{2} t \right)$$

$$\text{and } e^{\frac{\sqrt{\zeta^2 - 1}}{2} t} + e^{-\frac{\sqrt{\zeta^2 - 1}}{2} t} = 2 \cosh \left(\frac{\sqrt{\zeta^2 - 1}}{2} t \right)$$

$$Y(t) = 1 + \frac{s_b e^{s_a t} - s_a e^{s_b t}}{(s_a - s_b)} = 1 + \frac{s_b e^{s_a t} - s_a e^{s_b t}}{2 \frac{\sqrt{z^2 - 1}}{2}}$$

$$= 1 + \frac{e^{-\frac{z t}{\tau}}}{2 \frac{\sqrt{z^2 - 1}}{2}} \left(s_b e^{\sqrt{z^2 - 1} \frac{t}{\tau}} - s_a e^{-\sqrt{z^2 - 1} \frac{t}{\tau}} \right)$$

$$\text{Sub } \sqrt{z^2 - 1} = \lambda$$

$$= 1 + \frac{e^{-\frac{z t}{\tau}}}{2 \frac{\lambda}{\tau}} \left(s_b e^{\frac{\lambda t}{\tau}} - s_a e^{-\frac{\lambda t}{\tau}} \right)$$

$$= 1 + \frac{e^{-\frac{z t}{\tau}}}{2 \frac{\lambda}{\tau}} \left(\left(-\frac{z - \lambda}{\tau} \right) e^{\frac{\lambda t}{\tau}} - \left(-\frac{z + \lambda}{\tau} \right) e^{-\frac{\lambda t}{\tau}} \right)$$

$$= 1 + \frac{e^{-\frac{z t}{\tau}}}{2 \lambda} \left(-z e^{\frac{\lambda t}{\tau}} - \lambda e^{\frac{\lambda t}{\tau}} + z e^{-\frac{\lambda t}{\tau}} - \lambda e^{-\frac{\lambda t}{\tau}} \right)$$

$$= 1 + \frac{e^{-\frac{\zeta t}{2}}}{2\lambda} \left(-\zeta \left(e^{\frac{\lambda t}{2}} - e^{-\frac{\lambda t}{2}} \right) - \lambda \left(e^{\frac{\lambda t}{2}} + e^{-\frac{\lambda t}{2}} \right) \right)$$

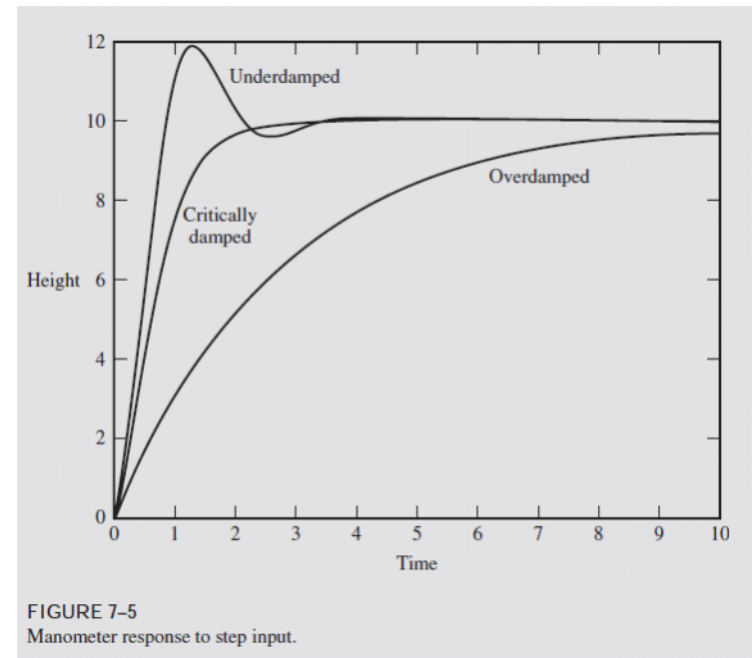
$$= 1 - e^{-\frac{\zeta t}{2}} \left(\left(\frac{e^{\frac{\lambda t}{2}} + e^{-\frac{\lambda t}{2}}}{2} \right) + \frac{\zeta}{\lambda} \left(\frac{e^{\frac{\lambda t}{2}} - e^{-\frac{\lambda t}{2}}}{2} \right) \right)$$

$$= 1 - e^{-\frac{\zeta t}{2}} \left(\cosh\left(\frac{\lambda t}{2}\right) + \frac{\zeta}{\lambda} \sinh\left(\frac{\lambda t}{2}\right) \right)$$

$$y(t) = 1 - e^{-\frac{\zeta t}{2}} \left[\cosh \sqrt{\zeta^2 - 1} \frac{t}{2} + \frac{\zeta}{\sqrt{\zeta^2 - 1}} \sinh \sqrt{\zeta^2 - 1} \frac{t}{2} \right]$$

Response becomes sluggish as ζ increases

$$y = 1 \text{ at } \infty t$$



$\text{Overshoot} = A/B$
 $\downarrow$
 increases
 for
 decreasing
 ζ

$$\text{Decay Ratio} = C/A = \text{overshoot}^2$$

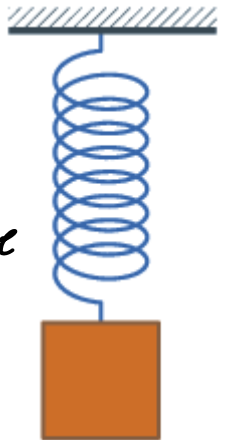
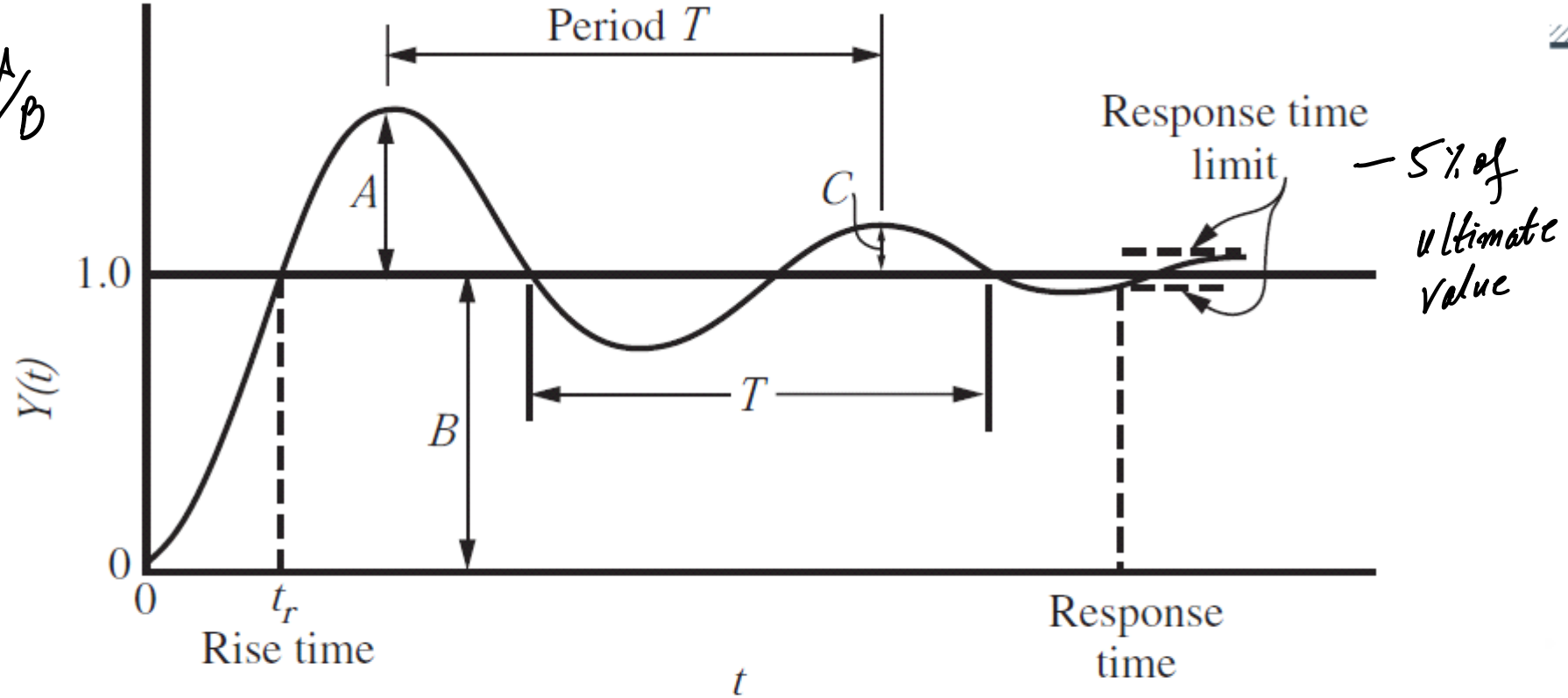


FIGURE 7-6

Terms used to describe an underdamped second-order response.

Overshoot $\rightarrow$ How much response exceeds the ultimate value following a step change

$$= \exp\left(\frac{-\pi \zeta}{\sqrt{1-\zeta^2}}\right)$$

With

$$y(t) = 1 - e^{-\frac{\zeta t}{\tau}} \cos\left(\frac{\sqrt{1-\zeta^2}}{\tau} t\right) - \frac{\zeta}{\sqrt{1-\zeta^2}} e^{-\frac{\zeta t}{\tau}} \sin\left(\frac{\sqrt{1-\zeta^2}}{\tau} t\right)$$

First peak occurs when $\frac{dy}{dt} = 0$ $\omega_d = \frac{\sqrt{1-\zeta^2}}{\tau}$

$$\frac{dy}{dt} = e^{-\frac{\zeta t}{\tau}} \cdot \frac{\sqrt{1-\zeta^2} + \zeta^2}{\tau \sqrt{1-\zeta^2}} \sin\left(\frac{\sqrt{1-\zeta^2}}{\tau} t\right) = 0$$

$$e^{-\frac{\zeta t}{\tau}} \neq 0 \quad ; \quad \sin\left(\frac{\sqrt{1-\zeta^2}}{\tau} t\right) = 0$$

$$\sin\left(\frac{\sqrt{1-\zeta^2}}{\tau} t\right) = 0$$

$$\frac{\sqrt{1-\zeta^2}}{\tau} t = n\pi$$

$$t_{peak} = \frac{n\pi\tau}{\sqrt{1-\zeta^2}}$$

Sub t_{peak} into $y(t)$

$$y_{peak} = 1 - e^{-\frac{\zeta}{\tau} \frac{n\pi\tau}{\sqrt{1-\zeta^2}}} \left[\cos\left(\frac{\sqrt{1-\zeta^2}}{\tau} \frac{n\pi\tau}{\sqrt{1-\zeta^2}}\right) + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin\left(\frac{\sqrt{1-\zeta^2}}{\tau} \frac{n\pi\tau}{\sqrt{1-\zeta^2}}\right) \right]$$

$$y_{peak} = 1 - e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \left[\underbrace{\cos \pi}_{-1} + \frac{\zeta}{\sqrt{1-\zeta^2}} \underbrace{\sin \pi}_0 \right]$$

$$y_{peak} = 1 - e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} (-1) = 1 + e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}$$

$$Y_{\text{peak}} - Y_{\text{final}} = M_p = \text{Overshoot}$$

for unit step func $\Rightarrow Y_{\text{final}} = 1$

$$\text{Overshoot} = Y_{\text{peak}} - 1$$

$$= e^{-\frac{\pi \zeta}{\sqrt{1-\zeta^2}}}$$

e.g. for step change of mag. 4

$$\frac{Y(s)}{X(s)} = \frac{10}{s^2 + 1.6s + 4}$$

Overshoot = ? need to find ζ

$$Y(s) = \frac{4}{s} \frac{10}{s^2 + 1.6s + 4}$$

$\Rightarrow$ general form

$$\hookrightarrow \frac{10/4}{\frac{1}{4}s^2 + \frac{1.6}{4}s + 1} = \frac{2.5}{0.25s^2 + 0.4s + 1}$$

$$\hookrightarrow \frac{2.5}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$2\zeta\omega_n = 0.4$$

$$\text{and } \omega_n^2 = 0.25 \Rightarrow \omega_n = 0.5$$

$$\Rightarrow \zeta = \frac{0.4}{2 \cdot 0.5} = 0.4$$

$$\text{Overshoot} = e^{-\frac{\pi \zeta}{\sqrt{1-\zeta^2}}}$$

$$= 0.2538$$

$$\% \text{ Overshoot} = 25.38\%$$

Find ultimate value of $y(t)$

$$\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} s Y(s) = \lim_{s \rightarrow 0} \frac{40s}{s(s^2 + 1.6s + 4)} = \frac{40}{4} = 10 = B$$

$$\begin{aligned} \text{Max value} &= B + A \\ &= 10 + 10(0.253) \\ &= 12.54 \end{aligned}$$

Period of oscillation
↗ coefficient of time in sin term of is radian frequency *coefficient of time*

$$y(t) = 1 - e^{-\frac{\zeta t}{\tau}} \cos\left(\frac{\sqrt{1-\zeta^2}}{\tau} t\right) - \frac{\zeta}{\sqrt{1-\zeta^2}} e^{-\frac{\zeta t}{\tau}} \sin\left(\frac{\sqrt{1-\zeta^2}}{\tau} t\right)$$

$$\omega = \frac{\sqrt{1-\zeta^2}}{\tau} = \text{radian freq}$$

$$\text{cyclical freq } f = \frac{\omega}{2\pi}$$

$$f = \frac{\sqrt{1-\zeta^2}}{2\pi\tau} = \frac{1}{T} \rightarrow \text{period}$$

$$T = \frac{2\pi\tau}{\sqrt{1-\zeta^2}} = \frac{2\pi \cdot 0.5}{\sqrt{1-0.4^2}} = 3.427$$

Rise time $t_r = ?$

From table

$$\frac{1}{s(\tau^2 s^2 + 2\zeta\tau s + 1)}$$

$\xrightarrow{t}$

$$1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\frac{\zeta t}{\tau}} \sin\left(\frac{\sqrt{1-\zeta^2}}{\tau} t + \phi\right)$$

$$y(t) = B \left[1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\frac{\zeta t}{\tau}} \sin\left(\frac{\sqrt{1-\zeta^2}}{\tau} t + \phi\right) \right]$$

$\downarrow$
 10

$$\phi = \tan^{-1}\left(\frac{\sqrt{1-\zeta^2}}{\zeta}\right)$$

For rise time $y(t) = 10$

$$\Rightarrow 10 = 10 \left[1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\frac{\zeta t_r}{\tau}} \sin\left(\frac{\sqrt{1-\zeta^2}}{\tau} t_r + \phi\right) \right]$$

$\zeta = 0.4, \tau = 0.5$

$$10 = 10 \left[1 - 1.091 e^{-0.8 t_r} \sin(1.833 t_r + 80^\circ) \right]$$

$$0 = e^{-0.8 t_r} \sin(1.833 t_r + 1.159)$$

↳ radians

$$e^{-0.8 t_r} \neq 0$$

$$\sin(1.833 t_r + 1.159) = 0$$

$$1.833 t_r + 1.159 = \sin^{-1}(0) = n\pi$$

$$t_r = \frac{3.14 - 1.15}{1.833} = 1.082$$